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Boundary-spanning search and sustainable competitive advantage: The mediating roles of exploratory and exploitative innovations

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ABSTRACT

Although the critical role of boundary-spanning search in facilitating sustainable competitive advantage (SCA) has been recognized, how boundary-spanning search affects SCA under the conditions that competitors also search remains unclear. By considering competition, this study divides boundary-spanning search into proactive and responsive searches, and develops a theoretical model, in which proactive and responsive searches respectively influence exploratory and exploitative innovations, and subsequently enhance SCA. Empirical results from Chinese advanced manufacturing firms show that proactive and responsive searches have inverted U-shaped effects on SCA. Additionally, when the levels of proactive and responsive searches are low and moderate, exploratory innovation plays a complementary mediating role between proactive search and SCA, whereas exploitative innovation plays a complementary mediating role between responsive search and SCA. When considering competition, this study offers a crucial condition under which exploratory and exploitative innovations can transform the efforts of boundary-spanning search to SCA.

1. Introduction

It is well known that sustainable competitive advantage (SCA) plays a critical role in helping firms to gain excellent performance. According to resource-based view, resources are the primary sources of SCA. Hence, in order to achieve success, firms should acquire resources, especially kinds of knowledge resources (Mahdi, Nassar, & Almsafir, 2019). Boundary-spanning search, as an important way to acquire external knowledge, refers to the process that firms search new knowledge across organizational and technological boundaries. It can bring in novel and heterogeneous knowledge to accelerate technological advancements and prevent competitors' imitation. Therefore, boundary-spanning search has become a pivotal driver of firms' sustainability performance (Jung & Lee, 2016; Wang, Xue, & Yang, 2020).

As Greve and Taylor (2000) argued, firms search knowledge in the environment where competitors also search. In the context of dynamic competition, firms generally take competitors' actions into account when conducting search activities (Tang, Fu, & Yang, 2019). Accordingly, competitors' searches may influence firms' search strategies and outcomes. When considering competition, boundary-spanning search can be divided into proactive search and responsive search (Katila & Chen, 2008; O'Cass, Heirati, & Ngo, 2014). Specifically, proactive search is the process that firms search external knowledge ahead of competitors to proactively cope with the changing environment; responsive search describes the process that firms search external knowledge by following competitors to responsively adapt to the changing environment.

Although the influence of competitors' searches on firms' searches has been recognized, how boundary-spanning search affects SCA under the conditions that competitors also search remains poorly understood. On the one hand, the main ideas of boundary-spanning search were developed in terms of firms' own experiences and past choices, such as search depth and search breadth (Laursen & Salter, 2006; Wang et al., 2020), and neglected the fact that competitors also search. The few studies that incorporate competition have inconsistent arguments on the impacts of proactive and responsive searches. For instance, Katila and Chen (2008) argued that proactive and responsive searches are positively related to firms' competitive advantage, though in different ways. However, other research argued that the returns of proactive and responsive searches are diminishing. As Baum, Li, and Usher (2000) suggested, proactive search is costly and time consuming, as early movers' searches are generally associated with risks and uncertainties. Meanwhile, Lieberman and Asaba (2006) demonstrated that responsive search raises the odds of core rigidity and cognitive myopia.

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